

Autonomous Theory Construction Laboratory

Lecture 12: Peer Comparison

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Overview

Peer systems are compared under a sealed configuration and a fixed reproduction protocol.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$rank = (L_R, N_{fail}, K, C, 1 - p_{replay})$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$config_{seal} = H(grammar, acceptor, seed, budget)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$p_{replay} = N_{match}/N_{check}$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- PRT-26 rules
- Peer Comparison Protocol
- MAS-54 reviewer appendix